

SDHD

$\Sigma\Delta$ -ADC

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Agenda

- o Excerpt: Analog Devices AN-283
- o Internal circuitry
- o Cypress SD-ADC

NYQUIST SAMPLING WITH ANALOG LOWPASS FILTER

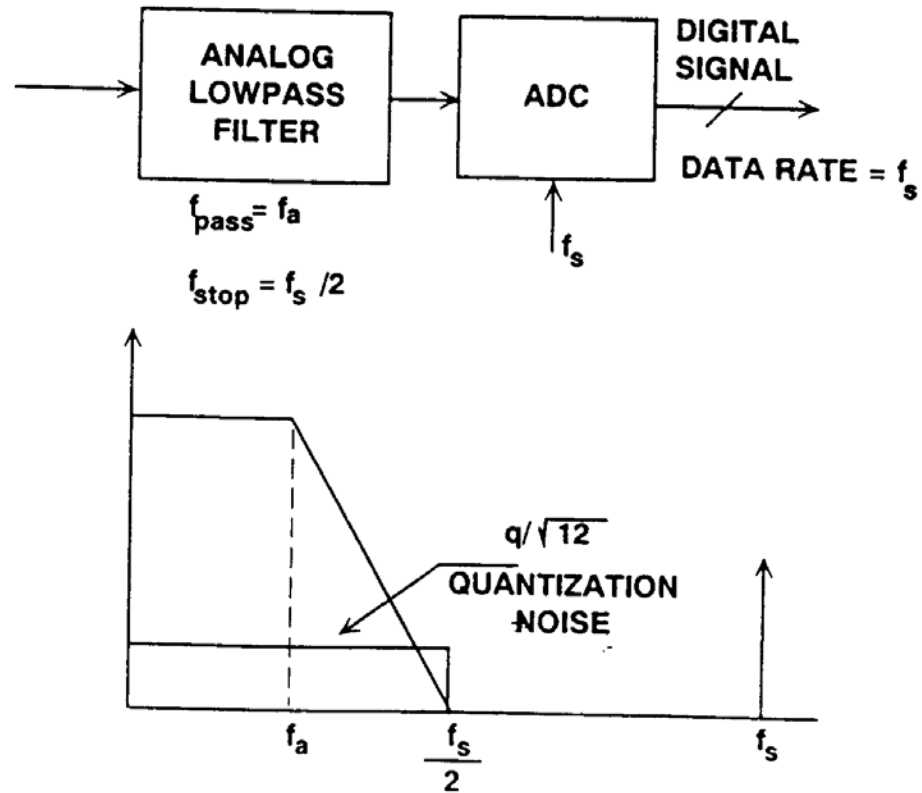
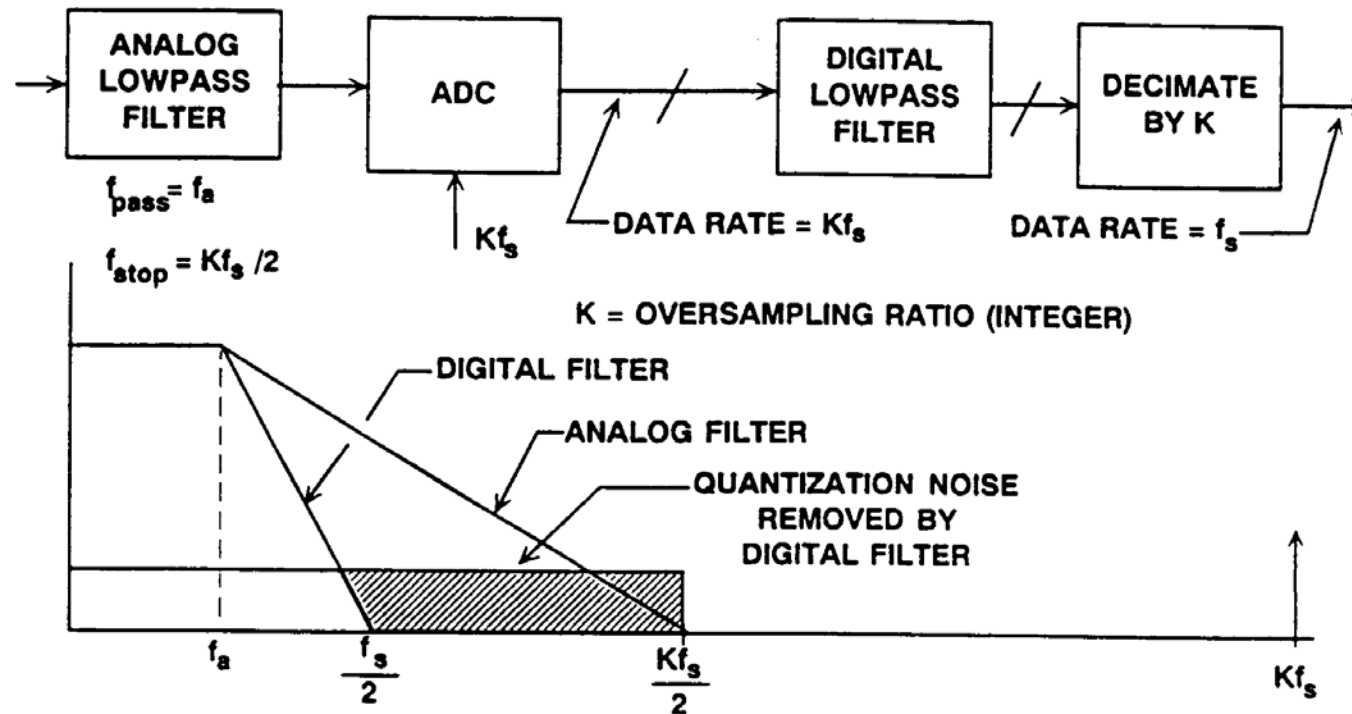
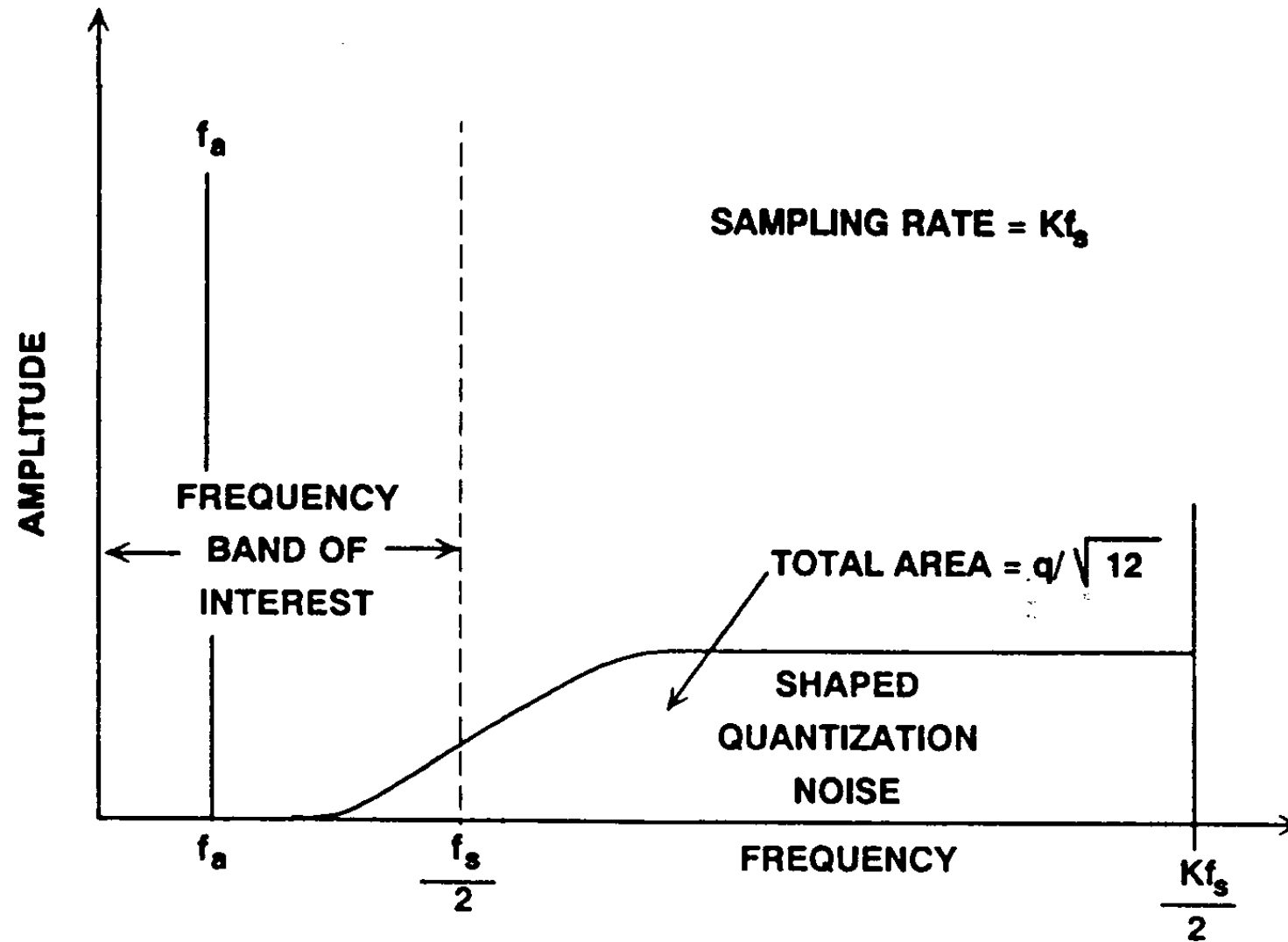


Figure 6.2

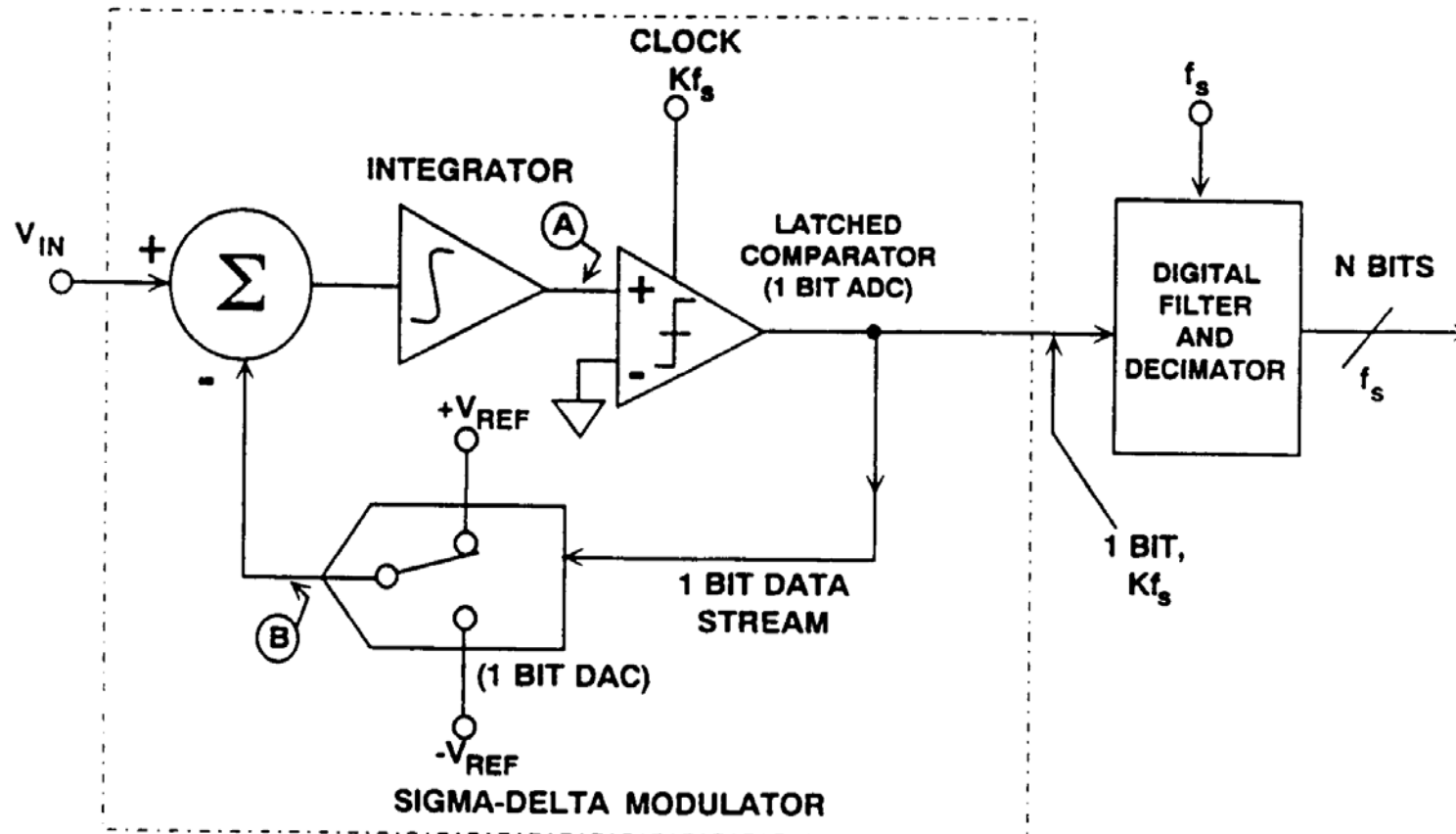
Assumption
Quantization is AWGN

OVERSAMPLING WITH ANALOG AND DIGITAL FILTERING





FIRST-ORDER SIGMA-DELTA ADC



FIRST AND SECOND-ORDER NOISE SHAPING FUNCTIONS

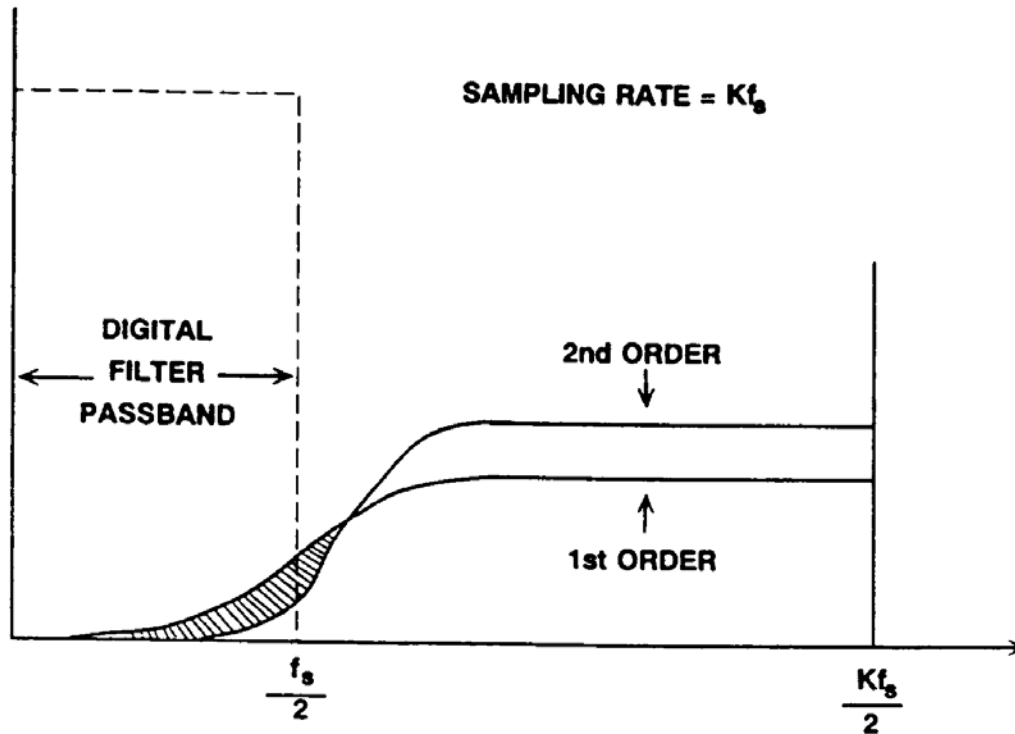
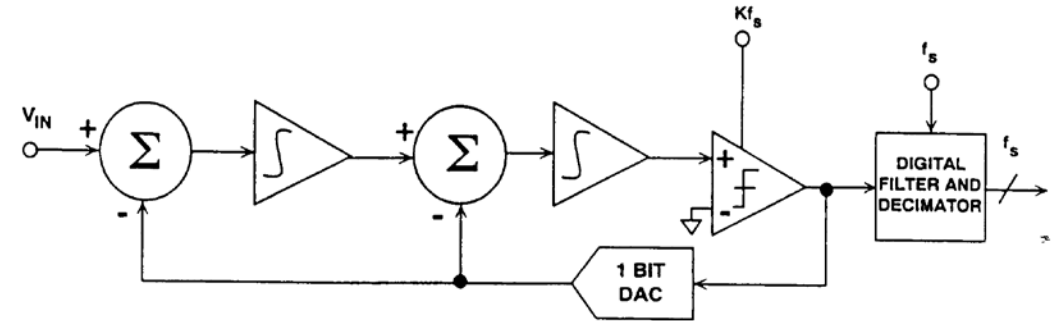
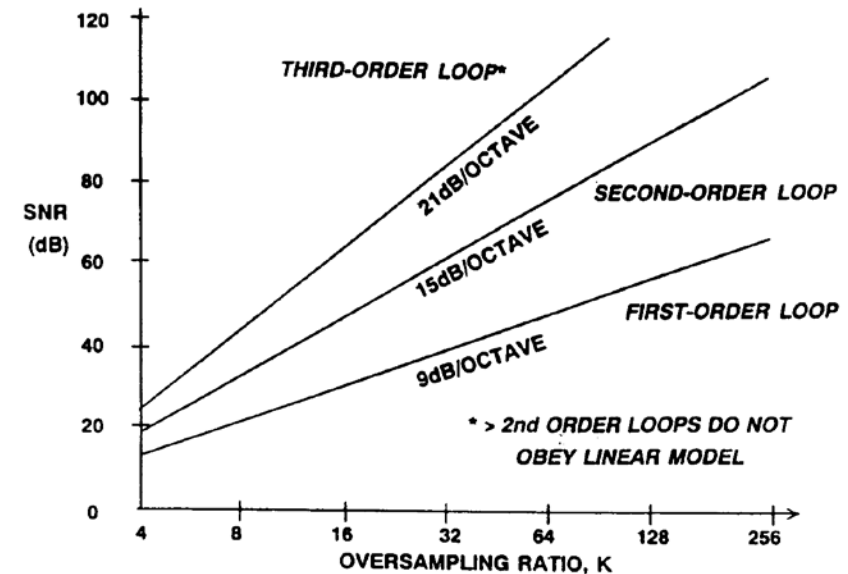
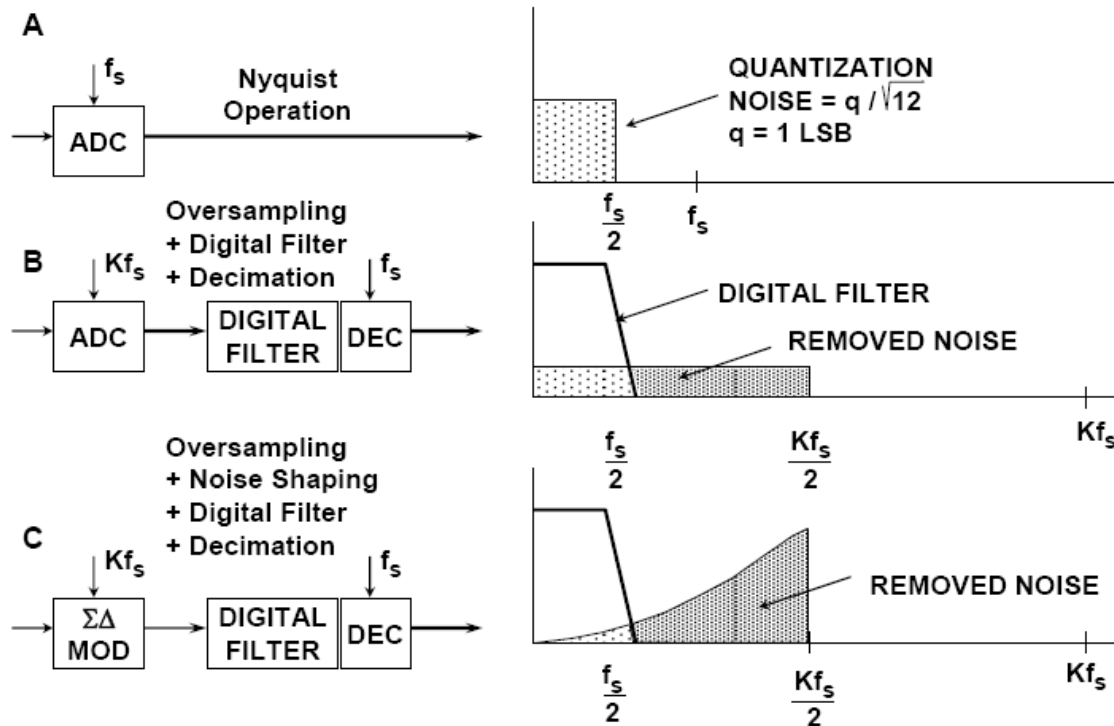


Figure 6.9

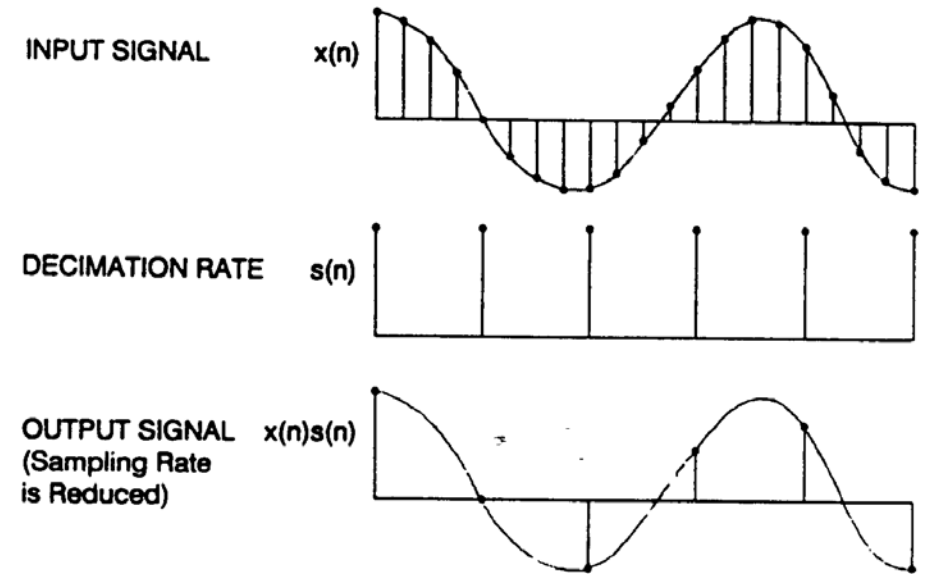


SNR VERSUS OVERSAMPLING RATIO FOR FIRST, SECOND, AND THIRD-ORDER LOOPS

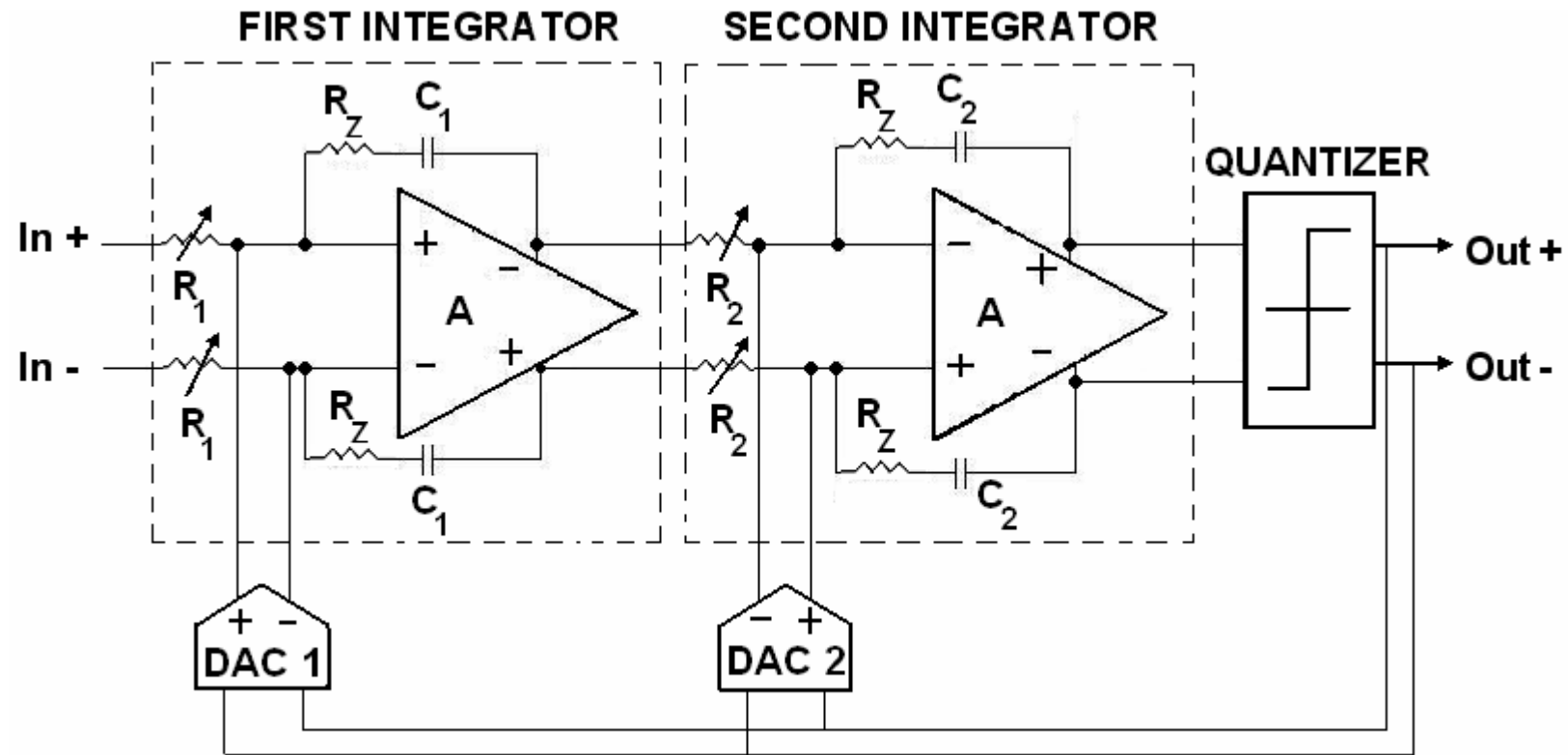




Decimation







Literature

- Application Note: AN-283 „Sigma Delta ADCs and DACs“ Analog Devices
- Tutorial: „ADC Architectures III: Sigma-Delta ADC Basics “ Walt Kester, Analog Devices
- Book: „Understanding Delta-Sigma Data Converters“ R. Schreier, G.C. Temes, Wiley, ISBN 0-471-46585-2, 2005